

LightLLM: A Versatile Large Language Model for Predictive Light Sensing

Jiawei Hu
University of New South Wales
Australia
jiawei.hu@unsw.edu.au

Hong Jia
University of Melbourne
Australia
hong.jia@unimelb.edu.au

Mahbub Hassan
University of New South Wales
Australia
mahbub.hassan@unsw.edu.au

Lina Yao
CSIRO
Australia
lina.yao@data61.csiro.au

Brano Kusy
CSIRO
Australia
brano.kusy@data61.csiro.au

Wen Hu
University of New South Wales
Australia
wen.hu@unsw.edu.au

ABSTRACT

We propose LightLLM, a model that fine tunes pre-trained large language models (LLMs) for light-based sensing tasks. It integrates a sensor data encoder to extract key features, a contextual prompt to provide environmental information, and a fusion layer to combine these inputs into a unified representation. This combined input is then processed by the pre-trained LLM, which remains frozen while being fine-tuned through the addition of lightweight, trainable components, allowing the model to adapt to new tasks without altering its original parameters. This approach enables flexible adaptation of LLM to specialized light sensing tasks with minimal computational overhead and retraining effort. We have implemented LightLLM for three light sensing tasks: light-based localization, outdoor solar forecasting, and indoor solar estimation. Using real-world experimental datasets, we demonstrate that LightLLM significantly outperforms state-of-the-art methods, achieving 4.4x improvement in localization accuracy and 3.4x improvement in indoor solar estimation when tested in previously unseen environments. We further demonstrate that LightLLM outperforms ChatGPT-4 with direct prompting, highlighting the advantages of LightLLM's specialized architecture for sensor data fusion with textual prompts.

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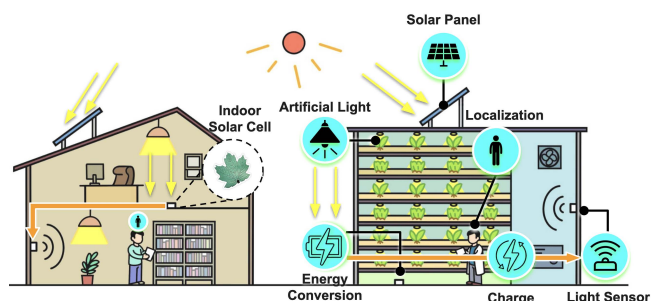


Figure 1: Illustration of multi-task PLS systems in the smart building (left) [47] and vertical farming (right) [6] systems. Outdoor solar panels harness natural energy for the overall system, while indoor solar cells power self-sustaining sensors used for indoor sensing and localization.

1 INTRODUCTION

Predictive light sensing (PLS) is becoming increasingly significant as the demand for intelligent and sustainable systems grows. For example, PLS can infer accurate indoor position using light spectral information (LSI) in smart buildings where GPS is unreliable, leveraging light sensor signatures to infer individual locations (see Figure 1 Left). Moreover, outdoor solar forecasting [21] is becoming increasingly critical for predicting the future power generation of rooftop solar panels based on weather conditions and sunlight availability. Similarly, indoor solar estimation [7] focuses on predicting potential solar output at various locations within an indoor space. For example, in settings such as vertical farms (see Figure 1 Right) [6]. These applications are vital as buildings and smart devices shift towards sustainability.

Traditional approaches to solving PLS tasks rely on task-specific deep learning models tailored to each individual application [60]. These models require separate training, manual tuning, and frequent retraining to adapt to changing data distributions, leading to significant engineering overhead. Furthermore, they often struggle to generalize to unseen

data or new environments [39], which is particularly problematic in real-world scenarios where domain shifts due to changing conditions are common. Techniques like data augmentation [16] and transfer learning [61] are often employed to mitigate this issue. However, generating high-quality synthetic data is resource-intensive, and transfer learning is limited when the target task diverges significantly from the source domain. Consequently, these models remain both computationally expensive and inflexible, restricting their scalability across diverse PLS tasks.

Large Language Models (LLMs), with their vast pre-trained knowledge and cross-domain generalization capabilities, present new opportunities to address the challenges outlined above. Inspired by the success of models like ChatGPT [35], which have demonstrated exceptional performance in reasoning and answering complex questions across diverse contexts in natural language processing, LLMs show promise for offering a unified solution across various IoT applications, including light sensing tasks. PLS tasks require integrating sensor data with contextual environmental information, a multimodal challenge that LLMs are well equipped to handle. Additionally, PLS data exhibit sequential dependencies and inherent noise, characteristics similar to natural language [30], making them well-suited for LLMs, which excel at processing structured sequences and extracting meaningful representations.

However, simply using LLMs with textual prompts, as explored in recent studies [23, 55], is insufficient for these tasks. LLMs are not inherently designed to process sensor data, such as light spectral information or time-series solar output, nor can they generate non-text outputs required for specialized applications like location classification or energy prediction. While recent efforts have extended LLMs beyond traditional language tasks—such as NetLLM [52] for networking and MentalLLM [54] for mental health assessment—these approaches still lack the ability to capture the domain-specific spatial, spectral, and environmental nuances critical for effective PLS sensing across different scenarios and environments.

We propose LightLLM, an innovative model designed to tackle light-based sensing tasks by leveraging the strengths of a pre-trained LLM. It integrates three key components: a task-specific sensor data encoder that extracts meaningful features from sensor inputs, a task-specific natural language prompt that provides contextual information about the environment, and a fusion layer that combines both sensor features and contextual prompts into a unified feature vector. This fused representation is then processed by the pre-trained LLM, which remains entirely frozen but fine-tuned using the LoRA (Low-Rank Adaptation) method. This unique architecture allows LightLLM to efficiently utilize the pre-existing general knowledge of LLMs while seamlessly

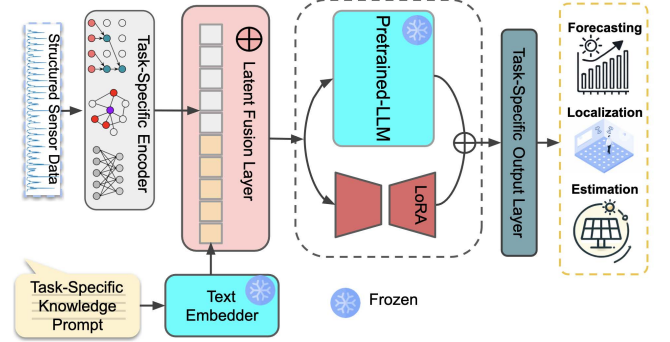


Figure 2: Architecture of LightLLM.

integrating domain-specific sensor data, enabling it to handle specialized light sensing tasks with high adaptability and minimal retraining effort.

Key contributions of this paper can be summarized as follows:

- We present LightLLM, a general-purpose framework for adapting LLM to a wide range of light-based tasks with minimal effort. To our knowledge, this is the first attempt to use LLM for PLS.
- We conduct comprehensive real-world experiments to validate the effectiveness of LightLLM for three distinct PLS tasks, light-based indoor localization, outdoor solar forecasting, and indoor solar estimation. Our results show the superior performance of LightLLM in terms of accuracy and generalization compared to state-of-the-art methods.
- We provide an extensive ablation study to evaluate the impact of each component within LightLLM, offering insights into its adaptability and performance across different tasks.

2 METHODOLOGY

2.1 System Overview

Our proposed framework, **LightLLM**, integrates advanced encoders for various data modalities with the capabilities of LLMs to address the challenges of diverse tasks. The framework, as illustrated in Figure 2, comprises five key components as followed.

Task-Specific Encoders: Tailored to individual tasks, such as Graph Neural Networks (GNNs) [53] for localization, Temporal Convolutional Networks (TCNs) [26] for forecasting, and Convolutional Neural Networks (CNNs) [28] for energy estimation. For tasks like indoor localization, a custom Knowledge Graph (KG) is integrated to represent a richer context, enabling improved interpretation of sensor data.

Task-Specific Knowledge Prompts: To provide structured contextual information, LightLLM employs task-specific prompts that encode domain knowledge essential to each task. These prompts are embedded as textual inputs,

aligning sensor data properties and environmental factors with the LLM’s knowledge base.

Latent Fusion Layer (LFL): A specialized component that integrates the encoded data with latent representations from the LLM. Task-specific prompts are leveraged to enhance the knowledge representation of sensor data.

LoRA: An efficient fine-tuning mechanism that adapts the LLM to specific tasks with minimal computational overhead.

Output Projection Layer: The final layer that projects the combined representations into the desired output format, ensuring compatibility with task-specific requirements.

2.2 Knowledge Graph for Enhanced Data Representation

In LightLLM, the custom KG is primarily designed for LSI indoor localization, where multiple light sources, and occlusions create complex spatial dependencies. However, for tasks such as outdoor solar forecasting, where the sun is the dominant and uniform light source, spatial dependencies between sensors are minimal, and simpler sequential models are sufficient. By focusing on relevant sensor interactions, the KG enhances the interpretability and efficiency of GNN-based encoders by concentrating on task-specific correlations.

KGs have demonstrated their effectiveness in representing structured data and improving task performance by organizing information into relational formats [8]. Existing ontologies like Brick [5] provide schemas for smart building metadata but lack the granularity needed for LSI indoor localization, which requires spatial relationships informed by sensor orientation, light angles, and occlusions. To address this, we design a task-specific KG that captures these dependencies to enhance performance.

The KG consists of nodes representing sensors and light sources, each with properties such as coordinates (x, y, z) , orientation, and detection range. Edges encode meaningful relationships based on distance, overlapping fields of view (FOVs), and obstacle-free pathways, ensuring efficient and interpretable connections. The detailed *Node Connection Algorithm* (Algorithm 1) constructs the KG in two phases: In phase 1, If two sensors are within a distance threshold and their FOVs overlap without obstruction, a “correlated” edge is added. In phase 2, If a sensor is within a valid horizontal and vertical range of a light source with no obstacles, a “light affect” edge is created.

Figure 3 illustrates an example KG: Sensor 3 is influenced by both Light Sources 1 and 2, while Sensors 1 and 2 are influenced solely by Light source 1. Since Sensors 2 and 3 are spatially correlated, any change in illumination—such as movement by occupants—is likely to affect readings from both sensors. An obstacle positioned between Sensor 1 and

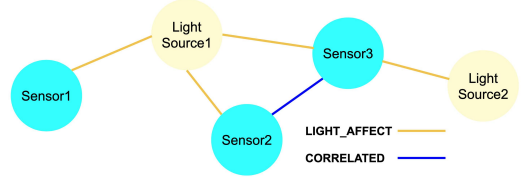


Figure 3: A simplified example of a KG illustrating sensor and light source interactions.

Sensors 2/3 blocks any direct correlation between them. Such KG is generated for each deployment based on the specific sensor layout and its spatial relationships with light sources.

By prioritizing obstacle-free, meaningful connections, our KG reduces complexity while ensuring realistic spatial modeling. This enhances LightLLM’s ability to interpret sensor data efficiently.

Algorithm 1: Node Connection Algorithm

Data: Set of sensors $S = \{s_1, s_2, \dots, s_n\}$, set of light sources $L = \{l_1, l_2, \dots, l_m\}$, set of obstacles O , field of view angle $angle$, distance threshold $distance_threshold$, detection range $range$, vertical threshold $vertical_threshold$

Result: Set of connections between sensors and light sources

```

1  $connections \leftarrow \{\}$ ;
2 for each pair of sensors  $(s_i, s_j)$  where  $i \neq j$  do
3    $d \leftarrow \text{Distance}(s_i, s_j)$ ;
4   if  $d \leq distance\_threshold$  then
5      $fov_i \leftarrow \text{CalculateFOV}(s_i, angle, range)$ ;
6      $fov_j \leftarrow \text{CalculateFOV}(s_j, angle, range)$ ;
7     if  $\text{FOVOverlap}(fov_i, fov_j)$  then
8       if  $\text{NoObstacle}(s_i, s_j, O)$  then
9         Append  $(s_i, s_j, \text{CORRELATED})$  to  $connections$ ;
10 for each sensor  $s_i$  and light source  $l_k$  do
11    $d_h \leftarrow \text{HorizontalDistance}(s_i, l_k)$ ;
12    $d_v \leftarrow \text{VerticalDifference}(s_i, l_k)$ ;
13   if  $d_h \leq distance\_threshold$  and  $d_v \leq vertical\_threshold$  then
14     if  $\text{NoObstacle}(s_i, l_k, O)$  then
15       Append  $(s_i, l_k, \text{LIGHT\_AFFECT})$  to  $connections$ ;
16 return  $connections$ ;

```

2.3 Task-Specific Encoders

The Task-Specific Encoders (TSEs) in LightLLM are designed to extract meaningful representations from diverse types of input data, ensuring that the most appropriate encoding techniques are applied.

2.3.1 Localization Encoder: Graph Neural Network For indoor localization, we employ a GNN to interpret spatial relationships among light sensors. The GNN leverages a knowledge graph structure where is constructed from sensor data using our Algorithm 1, where nodes represent sensor readings and edges capture spatial relationships (see Section 2.2). By aggregating information from neighboring nodes, the

GNN captures inter-sensor dependencies, enhancing spatial awareness and improving localization accuracy.

2.3.2 Forecasting Encoder: Temporal Convolutional Network

For the task of **solar energy forecasting**, we employ a TCN to handle the time-series data from outdoor solar cells. TCNs are ideal for capturing long-range dependencies in time-series data due to their architecture, which combines causal convolutions and dilation.

2.3.3 Energy Estimation Encoder: Convolutional Neural Network

To estimate **indoor solar energy output**, we use a CNN to analyze spectral sensor data. The CNN extracts hierarchical features across different wavelengths, helping identify both low-level and high-level patterns in the spectral data. This capability is critical for modeling the complex interactions between light spectra and solar cell energy output.

2.4 Task-Specific Knowledge Prompts

For each task, a **task-specific prompt** is generated to provide additional contextual knowledge. These prompts embed domain-specific features such as spatial information for localization or environmental conditions for solar forecasting. Unlike direct prompt-based reasoning (Section 3.4.1), which relies on an LLM’s dynamic response generation, task-specific knowledge prompts in LightLLM serve as static contextual embeddings that provide structured sensor-related background information to the model. These prompts do not function as iterative reasoning mechanisms but rather as an augmentation of the encoded sensor input. This added context helps guide the model in understanding the relationship between input features.

As shown in Figure 4, each task-specific prompt follows a structured template, which includes: **Dataset Description** outlines dataset characteristics, including data type (e.g., spectral wavelengths) and relevant factors. **Task Description** provides a brief explanation of the task. **Data Organization** specifies how the data is structured to help the model interpret input formats. **Key Input Characteristics** highlights statistical properties or patterns, such as temporal dependencies or spatial relationships.

This structure ensures that LightLLM can effectively address the unique challenges of each task, particularly when operating in unseen environments where sensor layout or lighting conditions differ from the training data. By combining the LLM’s generalized knowledge of spatial relationships and environmental patterns with the prompt’s specific spatial configurations, LightLLM gains the ability to reason about relationships between sensor placement, light source distribution, and sensor readings, leading to improved inference accuracy and overall system performance.

These prompts are then embedded using the LLM’s text embedding mechanism, and the resulting embedding objects are integrated into the Latent Fusion Layer.

<Dataset Description>: The Power Output Measurements dataset contains data from a 30-kW rooftop photovoltaic array, with power output logged every 1 minute. The dataset captures the power generation patterns under varying sunlight and weather conditions.
<Task Description>: Forecast the next $\{T\}$ steps using the previous $\{P\}$ steps of historical data, accounting for variations in sunlight intensity and environmental factors.
<Data Organization>: $\{PV_output\}$.
<Key Input Characteristics>:

- Sunlight intensity over the past period, $\{min_values\}$ to $\{max_values\}$,
- Cloud cover and weather variability, impacting solar output trends,
- Temporal patterns such as day-night cycles,
- Geographic influence from location-specific sunlight exposure at $\{latitude\}$, $\{longitude\}$.

The focus is on modeling how geographic and environmental factors drive solar output variability for more accurate predictions.

Figure 4: Task-Specific Knowledge Prompt Example: Solar Energy Forecasting Prompt.

2.5 Adaptive Latent Fusion Layer

The LFL in LightLLM is a key component designed to integrate encoded features from task-specific encoders with prompt embeddings generated from task-specific knowledge prompts. This integration is achieved through a multi-head attention mechanism [45], allowing the model to leverage the strengths of both the feature encoders and the LLM, the structure is shown in Figure 5. The LFL ensures that the encoded data is aligned with the LLM’s latent space, facilitating effective fusion of multimodal information.

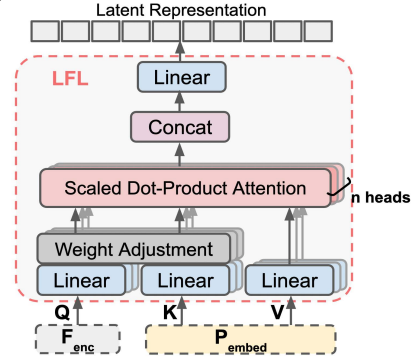


Figure 5: Overview of Latent Fusion Layer.

The **encoded features** F_{enc} from the task-specific encoder and the **embedded prompt** P_{embed} are fused using a multi-head attention mechanism within the LFL. Specifically, the query Q , key K , and value V matrices are computed from the encoded features and the prompt embeddings as follows:

$$Q = W_Q F_{enc}, \quad K = W_K P_{embed}, \quad V = W_V P_{embed}.$$

Here, the prompt embeddings P_{embed} encapsulate structured, task-specific knowledge that complements the encoded sensor features. By assigning P_{embed} as both the key

and value, the attention mechanism identifies relevant knowledge and integrates it into the encoded features F_{enc} . The key (K) assesses relevance, while the value (V) provides the information to enhance the final representation.

Each head in the multi-head attention mechanism works with different learned weight matrices W_Q, W_K and W_V . It splits the matrices into n_{heads} parallel subspace, computes attention separately for each head, and then concatenates the results.

In addition to the standard attention mechanism, the LFL incorporates a weight adjustment mechanism to control the contribution of the encoded data and the text embeddings. This is achieved through learnable parameters λ_T and λ_S , which adjust the weights of the target and source embeddings, respectively. The target embedding T' , which represents the encoded sensor features, and the source embedding S' , which represents the task-specific prompt information, are projected and scaled as follows:

$$T' = \lambda_T \cdot Q, \quad S' = \lambda_S \cdot K. \quad (1)$$

The attention scores are then computed using the scaled dot-product of the weighted embeddings:

$$\text{Scores} = \text{softmax} \left(\frac{T' S'^T}{\sqrt{d_k}} \right), \quad (2)$$

where S'^T denotes the transpose of the source embedding matrix and d_k the dimension of the key vectors. Once the fusion process is completed, the resulting representation is passed into the LLM to generate the reprogrammed latent representation. The resulting latent representation combines the strengths of the encoded sensor data and the prompt knowledge, providing a task-specific feature that is well-suited for downstream tasks in PLS.

The LFL also includes normalization and dropout layers to ensure stable training and prevent overfitting. LFL effectively leverages both pre-trained knowledge and task-specific information, allowing LightLLM to produce highly accurate, adaptable outputs across various PLS applications.

2.6 Low-Rank Adaptation

To efficiently adapt the pre-trained language model for our specific downstream tasks, we utilize LoRA [18], which introduces low-rank matrices around the original pre-trained LLM, allowing for dimensionality reduction and expansion, effectively performing a rank adaptation. During training, the parameters of the LLM remain frozen, while only the low-rank matrices A and B are trained. This significantly reduces the number of trainable parameters while preserving the model's capacity to adapt to new tasks.

Initially, the reduction matrix A is randomly initialized using a Gaussian distribution, while the expansion matrix B

is initialized as a zero matrix. This setup ensures that the pre-trained behavior of the LLM is maintained at the beginning of training, with the adaptation gradually introduced as the matrices are trained.

The modified weight matrix W' is computed as follows:

$$W' = W + \alpha AB, \quad (3)$$

where W is the pre-trained weight matrix (e.g., a frozen LLM). The dimensions of low-rank matrices $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times d}$ are significantly smaller than the full weight matrix $W \in \mathbb{R}^{d \times d}$, significantly reducing the number of parameters being fine-tuned.

During the forward pass, the input x is computed as follows:

$$W'x = Wx + \alpha(AB)x. \quad (4)$$

This approach is analogous to a residual connection, where the low-rank updates simulate the full fine-tuning process by introducing an adaptable layer on top of the pre-trained LLM. When the rank r equals the dimension of W , LoRA effectively performs full fine-tuning, but with $r \ll d$, LoRA provides a more computationally efficient way to adapt the model. In the inference phase, LoRA introduces minimal additional latency, with parallel computations for Wx and $(AB)x$.

In our model, we carefully chose the settings $r = 64$, $\alpha = 64$, and dropout of 0.1, to achieve an optimal balance between task-specific adaptation and preserving the pre-trained knowledge of the LLM, while maintaining computational efficiency. While the initial fine-tuning may benefit from cloud resources, our lightweight LoRA approach ensures that real-time inference and subsequent minor adaptations can be efficiently executed on edge devices, such as smart home hubs equipped with GPUs or high-performance CPUs [13]. This framework significantly reduces the cost and complexity of on-site deployment, making it feasible for smart building or smart home scenarios.

2.7 Task-Specific Output Layer

The output layer of LightLLM is designed to handle the specific requirements of different tasks by employing task-specific heads. This ensures that the final model outputs are appropriate for each application.

For example, for the localization task, the output layer is configured as a probability classification head [15]. This head outputs a probability distribution over possible locations, allowing the model to predict the most likely position based on the input data. The classification head uses a softmax activation function to convert the model's logits into probabilities, thereby facilitating accurate location predictions.

The flexible design of this task-specific head allows LightLLM to be easily adapted to different tasks. By incorporating the appropriate headers, the model is able to provide accurate, task-relevant output, whether it involves classification, regression, or other output formats. This module ensures that the model’s predictions are not only accurate but also customized to the unique characteristics of each PLS task.

3 EVALUATION

Implementation. The fine-tuning of LightLLM was conducted using an NVIDIA A100-80GB GPU, leveraging the PyTorch [38] framework. To ensure optimal performance, LightLLM was fine-tuned separately for each sensing task rather than trained as a unified model. Each task was handled independently to allow task-specific encoders and prompts to extract meaningful features. In our experiments, the system is built upon a pre-trained model derived from the Llama-7B[44]. The training process took 10-16 hours for each task model. The fine-tuning procedure involved several key steps: (1) Data preprocessing, where raw sensor data was cleaned, normalized, and transformed into appropriate formats for the encoder. (2) Task-specific prompting, where contextual knowledge relevant to each task was structured and embedded as input to the LLM. (3) Fine-tuning with LoRA, where only the low-rank adapters were trained while keeping the pre-trained LLM frozen. (4) Performance evaluation with validation datasets to ensure generalization. Our code and dataset are available at <https://github.com/KuuGary/LightLLM.git>.

3.1 Light Spectral Localization

Indoor localization using light spectral information has proven to be a highly effective technique for achieving passive Visible Light Positioning (VLP) in indoor environments. This method leverages the unique properties of light in different spaces to estimate the location of users. One of the leading systems for spectral-based localization is *Iris* [19], which employs a conditional Generative Adversarial Network (cGAN) for data augmentation to enhance its performance.

To evaluate the performance of LightLLM in indoor localization, we collected data from two distinct indoor environments: a large office ($108m^2$) illuminated by diffused ceiling lights and an apartment ($25m^2$) using directed LED downlights. Both environments contain standard furniture layouts, including desks, chairs, and partitions, as well as windows, ensuring realistic indoor settings. We strategically deployed 27 spectral sensors in the office and 17 in the apartment, to capture a diverse range of spectral interactions. Each environment was configured to collect spectral data across multiple sessions, including scenarios with and without natural light interference.

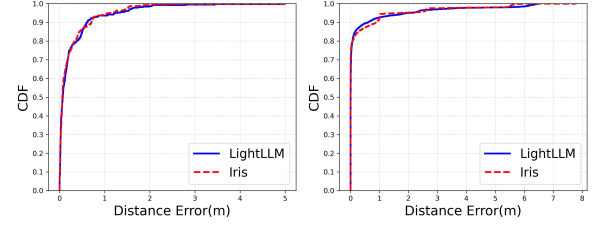


Figure 6: Localization error under *seen* scenarios: (Left) Apartment, (Right) Office.

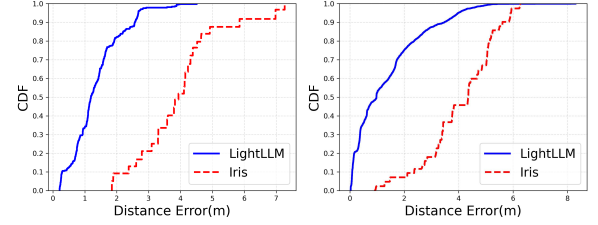


Figure 7: Localization error under *unseen* scenarios: (Left) Apartment, (Right) Office.

In the **seen environment** tests, where both training and testing data are collected from the same indoor room, LightLLM shows comparable or slightly better performance than *Iris* (see Figure 6). For example, in the 90th percentile error, where LightLLM achieves 0.608m vs 0.705m for *Iris* in the apartment environment, and LightLLM achieves 0.89m, while *Iris* performs slightly worse at 0.60m of 90th percentile error in the office environment.

The main benefit of LightLLM can be seen in the **unseen environment**, where the training data does not include the environment used in the test set (e.g., train with apartment’s dataset and test with office’s dataset). Traditional models like *Iris* do not generalize well in these scenarios. Consider Figure 7: *Iris* sees a dramatic increase in the median error of 3.93m and 4.35m for office and apartment environments, respectively. In contrast, LightLLM achieves a far better generalization with a median error of 0.98m and 1.19m for office and apartment, respectively, corresponding to a 4.4x improvement. *Iris* exhibits a significant drop in accuracy in unseen environments due to its reliance on a fingerprint-based learning approach, which requires extensive training data collected in the target environment. In contrast, LightLLM generalizes effectively by integrating spatial and contextual information through prompt knowledge injection, allowing it to adapt to different sensor distributions without retraining.

In summary, LightLLM exhibits competitive performance in seen environments but shows superior generalization capabilities in unseen environments. The ability of LightLLM to adapt to new environments with minimal degradation in performance highlights its advantage over traditional systems like *Iris*, which perform poorly in new environments. This makes LightLLM a more robust and scalable solution

for light spectral indoor localization, especially in real-world applications where environments are diverse and complex.

3.2 Outdoor Solar Energy Forecasting

Solar forecasting plays a crucial role in optimizing energy management systems and improving the reliability of renewable energy sources. Accurate forecasts of photovoltaic (PV) generation help ensure energy grid stability and enable more efficient integration of solar energy into the energy mix. To achieve these predictions, solar forecasting models typically rely on a variety of data models, including historical PV generation and environmental data, such as images of the sky that capture cloud movement.

The dataset used in this evaluation, SKIPP'D [33] includes over three years of PV power generation data. It covers a diverse range of weather conditions, providing an excellent opportunity to assess the performance of solar forecasting models across various scenarios.

To evaluate the performance of LightLLM, we employ several commonly used metrics in solar forecasting [14]: The **Continuous Ranked Probability Score (CRPS)** measures probabilistic prediction accuracy, where a lower CRPS indicates better performance. The **Forecast Skill (FS)** reflects the percentage improvement over a baseline method (e.g., smart persistence [9]), with higher FS values representing better improvements. The **Winkler Score (WS)** assesses the width of prediction intervals at a given probability level, favouring narrow intervals that still capture the true value, where a lower WS indicates a better probabilistic forecast. These metrics enable a comprehensive evaluation of forecasting performance under various conditions.

We compared LightLLM against several state-of-the-art methods, including traditional method ConvLSTM [42], and SkyGPT [34], a leading model that employs sky images in combination with U-Net for solar energy forecasting, and TimeLLM [24], which also utilizes LLM for time-series predictions. All of them are comparing with a commonly used baseline method, smart persistence model [9]. The results in the seen environment are shown in the Table 1. LightLLM outperforms SkyGPT in several key metrics, obtaining a CRPS of 1.92 kW, representing a 33.7% improvement in FS, which is significantly higher than SkyGPT's performance. Furthermore, LightLLM records a WS of 20.80, demonstrating superior accuracy in matching the predicted solar output distribution to the actual data. Additionally, while TimeLLM offers solid performance, LightLLM consistently delivers better results across all metrics, highlighting the advantage of integrating task-specific encoders and the LFL in LightLLM.

In the unseen scenario, LightLLM is evaluated on a completely separate time period of data that the model has not encountered during training, specifically, we trained on data from 2017 Mar to 2019 Oct and tested on data from 2019 Nov

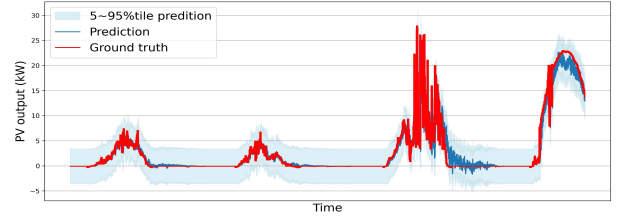


Figure 8: Prediction curve of LightLLM and ground truth with prediction intervals (5-95%) under unseen scenarios.

to 2019 Dec. It is intended to simulate real-world conditions to evaluate the final performance of the model. Consider Table 1 and Figure 8, LightLLM maintains its edge with a CRPS of 2.52 kW and a FS value of 31.4%, outperforming SkyGPT (CRPS: 2.81 kW, FS: 23.4%) and TimeLLM (CRPS: 2.67 kW, FS: 27.3%). Furthermore, LightLLM records a WS of 22.56, again lower than SkyGPT's 26.70 and TimeLLM's 24.30, demonstrating its robustness. This indicates that LightLLM not only excels in seen conditions but also generalizes well in unseen settings.

The comparison clearly shows that LightLLM offers better performance across both seen and unseen environments compared to other leading models such as SkyGPT and TimeLLM. This improvement is primarily due to the TSE introduced in Section 2.3, which effectively extracts temporal features from solar data, and the LFL detailed in Section 2.5, which enhances data representation with additional contextual descriptions. The results in Section 3.4.2 further confirm the impact of these components.

3.3 Indoor Solar Energy Estimation

3.3.1 Challenges in Real-World Solar Energy Estimation. The increasing use of indoor solar cells in low-power IoT devices [1] and interactive surfaces [32] presents new opportunities for energy harvesting. Given that different types of solar cells exhibit unique responses to varying light conditions, optimizing their placement and understanding their energy output in real-world environments becomes critical.

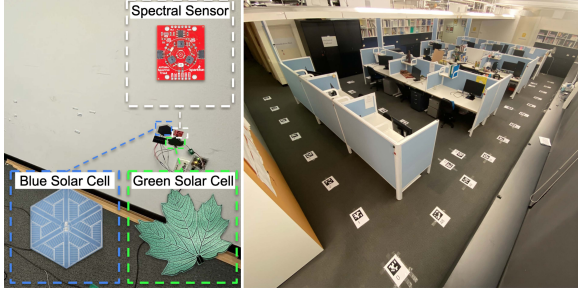
Fortunately, existing technology allows us to simulate and collect spectral sensor data [20]. However, the challenge lies in accurately translating this spectral data into solar energy output for each unique solar cell. While mathematical models exist to predict this conversion under idealized conditions, their accuracy diminishes in complex, real-world scenarios due to various discrepancies between theory and practice.

In theory, the short-circuit current density J_{sc} depends on the external quantum efficiency (EQE) and the incident photon intensity at each wavelength $P_{in}(\lambda)$ [51]. Following [31], the total current generated by a solar cell is given by:

$$\text{Current} = A \cdot \int_{\lambda_{min}}^{\lambda_{max}} a(\lambda) \cdot I(\lambda) \cdot \lambda \cdot d\lambda, \quad (5)$$

Table 1: LightLLM vs. state-of-the-art for outdoor solar forecasting, with Smart Persistence as the baseline.

Method	Seen			Unseen		
	CRPS[kW]↓	FS[%]↑	WS↓	CRPS[kW]↓	FS[%]↑	WS↓
Smart Persistence	2.89	0	-	3.67	0	-
ConvLSTM	2.70	6.6	41.12	3.09	15.8	43.82
SkyGPT	2.49	13.8	26.72	2.81	23.4	26.70
TimeLLM	2.27	21.6	21.30	2.67	27.3	24.30
LightLLM	1.92	33.7	20.80	2.52	31.4	22.56

**Figure 9: Experimental Pictures. (Left) Deployment of spectral sensors and semi-transparent solar cells with different absorption rates in a controlled setup. (Right) Real office environment used for evaluating LSI localization.**

given the spectral absorption rate $a(\lambda)$, the incident spectrum $I(\lambda)$, and the surface area A of the solar cell.

However, our real-world experiments revealed significant discrepancies between theoretical predictions and actual solar cell performance. In these tests, we deployed both spectral sensors (AS7265x) with 18 wavelength channels and solar cells with different absorption rates in an indoor room setup, as shown in Figure 9 (Left). All measurements were synchronized via an Arduino device. We compared actual solar cell outputs with theoretical predictions from Eq. (5), using Mean Absolute Percentage Error (MAPE) and Mean Squared Error (MSE):

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{\text{Actual}_i - \text{Predicted}_i}{\text{Actual}_i} \right| \times 100, \quad (6)$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\text{Actual}_i - \text{Predicted}_i)^2, \quad (7)$$

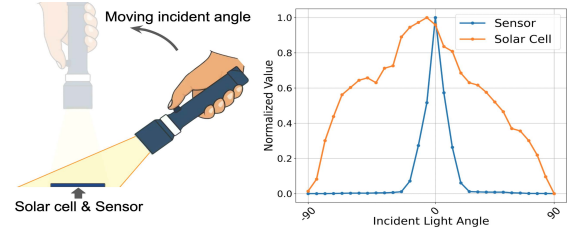
where n represents the total number of data samples, with $n = 1,035$ in our experiments.

Table 2 shows that theoretical models yield an average MAPE of **46.83%**, with individual errors of 44.31% (Blue Solar Cell) and 49.34% (Green Solar Cell). This high error rate underscores the limitations of the equation-based approach.

One key issue is the difference between spectral sensors and solar cells. Spectral sensors [4] have a defined FOV, whereas solar cells maximize absorption over their

Table 2: Actual vs. predicted photocurrent (mA) for solar cells at various indoor locations.

Location	Blue Solar Cell		Green Solar Cell	
	Actual	Predicted	Actual	Predicted
1	0.596	0.335	0.459	0.249
2	0.547	0.306	0.523	0.242
3	0.097	0.141	0.348	0.517
Avg MAPE[%]	44.31		49.34	

**Figure 10: Experiment results show the difference between the solar cell and light sensor readings when the light hits them from different angles.**

surface [37]. Solar cells are most efficient when light is perpendicular; as the incidence angle deviates, absorption decreases. Figure 10 illustrates this effect, showing significant differences in sensor and solar cell readings under varying angles.

Additional sources of error include variations in solar cell packaging, which affect light interaction, and manufacturing inconsistencies in absorption curve data. These challenges underscore the infeasibility of purely physics-based models for indoor solar energy estimation. Given the limitations of theoretical models in predicting solar cell performance under real-world conditions, we turn to data-driven approaches.

3.3.2 Performance Evaluation Results Under Seen Environment. We then tested several machine learning methods under a seen environment, where the training set contained data from the same environment as the test set. As summarized in Table 3. In this controlled setting, data-driven methods significantly outperformed the physical equation-based approach. Among the data-driven methods, Random Forest achieved the lowest MAPE of 2.58%, while LightLLM closely followed with a MAPE of 2.59%. In terms of MSE,

Table 3: Comparison of different machine learning methods vs. LightLLM for indoor solar estimation.

Method	Seen		Unseen	
	MSE	MAPE[%]	MSE	MAPE[%]
Linear Regression	19.40	6.15	3592.97	168.26
Random Forest	5.76	2.58	2438.28	140.91
Gradient Boosting	5.50	2.62	2465.54	141.64
Decision Tree	10.29	2.70	2934.94	154.20
SVR	15.96	5.41	2013.49	128.54
MLP	6.11	2.68	819.63	83.02
CNN	5.98	2.60	959.81	89.61
LightLLM	5.72	2.59	263.21	26.33

LightLLM achieved the best performance 5.72. This demonstrates that, given sufficient training data from the same environment, the data-driven models can effectively capture the relationship between spectral data and solar cell energy output.

Results Under Unseen Environment. To further test the generalization of the models, we evaluated them on unseen environments, where the test set was drawn from environments not present in the training set. We specifically altered the lighting and spatial configuration within the room. For example, we introduced two diffused standing lamps, and rearranged the room layout by changing the furniture. The results for this setting are shown in Table 3. Most machine learning models suffered a significant performance drop. For example, Linear Regression and Random Forest showed MAPE values exceeding 140%, illustrating their tendency to overfit and their lack of generalization capability. In contrast, LightLLM demonstrated remarkable robustness, achieving a MAPE of 26.33% and MSE of 263.21, the best among the compared methods. We note that LightLLM consistently outperforms the traditional equation-based approach in both seen and unseen environments.

3.4 Ablation Study

3.4.1 LightLLM vs. Direct Prompting To evaluate the effectiveness of LightLLM against a simpler method of directly prompting an LLM (e.g., ChatGPT-4), we firstly conducted an experiment using the indoor localization task as a representative. We compared both zero-shot and few-shot prompting approaches using models like GPT-4, GPT-3.5 [35], Llama 3 [44], and Mistral Large 2 [3]. Our goal was to determine whether carefully crafted prompts, including chain-of-thought (CoT) [48] prompting strategy, could compete with the task-specific design of LightLLM.

Prompting Setup. The ability of LLMs to understand and reason across a wide range of tasks makes them powerful, particularly when paired with well-constructed prompts. Recent studies [29] have demonstrated that proper prompt engineering can significantly improve LLM performance

on domain-specific tasks. Therefore, for the direct prompting method, we created a detailed prompt, leveraging CoT reasoning to improve its ability to process the sensor data step by step. A prompt example design is shown as in Figure 11. First, the **Instruction** gives a detailed description of the room setup, including the locations and orientations of spectral sensors, which is essential for the model to understand the spatial relationships in the environment. The **Data** section introduces the sensor readings and the changes in wavelength due to movement. The **Question** then prompts the LLM to determine the person’s location based on the given data, while the inclusion of a step-by-step analysis encourages logical reasoning, following the CoT prompting strategy.

Instruction: In an indoor room, illuminated by overhead lights, we have placed spectral sensors all around, facing inside the room. The location of spectral sensors are as follows: {*sensor locations* (x, y, z)}. Their orientation are as following: {*sensor orientations* (x, y, z)}. When someone passes by, the light changes, hence the readings on each wavelength of the spectral sensors also change. When the room is empty, the sensors’ readings are as follows: {*readings*}.

Data: The sensor data given in the room has 18 wavelengths. The change of 18 wavelengths’ reading comparing with the empty room for each sensor are given below: {*changes*}.

Question: The person’s location belongs to one of the following categories: {*locations* (x, y)}. Please tell me where the person is based on the given information. Make an analysis step by step.

Response: {*answer*}

Figure 11: Prompt example for LSI-based localization.

Zero-shot prompting. As shown in Table 4, GPT-4 achieved the best performance among different models, a median error of 1.77m for the LSI localization task. This error is considerably larger than the result achieved by LightLLM in the same scenario, where the median error was 0.98m. From the responses given by ChatGPT-4, we also found that, despite the detailed instruction and CoT prompt, ChatGPT-4 primarily used a very basic heuristic approach: it attempted to estimate the person’s location by selecting the coordinates closest to the sensors that detected the largest change in spectral readings.

To inject domain-specific context without additional training, researchers have explored **Few-shot prompting** by embedding a few examples directly within the prompt (e.g., [2]) while the LLM parameters without tuning. From the results in Table 4 we can observe, while this did improve the performance slightly (e.g., median error from 1.77m to 1.70m), the results still fell short compared to LightLLM.

Furthermore, we observed that certain models, such as Gemini 1.5 [41], refused to provide specific answers to queries about location coordinates. Instead, it offered approximate descriptions, such as “The estimated location based on the given sensor data is near the bottom-center of the room,” or stated, “I cannot provide a specific coordinate.”

The limitations of direct prompting were not confined to localization. We have also conducted the experiment for the

Table 4: Performance comparison of direct prompting vs. LightLLM across models and tasks.

Method	Model	Outdoor Solar Forecasting CRPS[kW]↓	Indoor Localization Median Error[m]↓	Indoor Solar Estimation MAPE[%]↓
Zero-shot Prompting	<i>GPT-4</i>	2.57	1.77	48.72
	<i>GPT-3.5</i>	2.96	2.03	45.35
	<i>Llama 3</i>	3.37	3.84	45.44
	<i>Mistral Large 2</i>	2.69	2.36	56.38
Few-shot Prompting	<i>GPT-4</i>	2.42	1.70	50.43
	<i>GPT-3.5</i>	2.87	1.92	45.04
	<i>Llama 3</i>	3.20	3.12	44.80
	<i>Mistral Large 2</i>	2.55	2.30	49.06
LightLLM	-	1.64	0.98	26.33

other two tasks. We used a smaller scale of the dataset to evaluate the performance of solar energy forecasting. The forecasting results by direct prompting were poor, leading to high CRPS values (e.g., GPT-4 zero-shot at 2.57 kW, compared to LightLLM’s 1.64 kW). Similarly, for indoor solar energy estimation, the prompt-based estimates deviated significantly from actual measurements, as the LLMs failed to model the complexities of spectral data conversion (e.g., Llama 3 at 45.44% vs LightLLM’s 26.33%). Interestingly, when we used few-shot prompting by providing some examples, the performance of certain models decreased slightly in indoor solar estimation task. This decline can be attributed to the fact that these examples may have misled the LLM models, causing them to overfit to specific patterns.

This ablation study shows that, LightLLM was able to handle the complexity of the sensor data much more effectively, demonstrating that for tasks requiring complex multimodal data interpretation, it is not sufficient to simply use LLM with direct prompting methods.

3.4.2 Impact of Key Components The performance of LightLLM is significantly enhanced by its core components: The inclusion of prompt knowledge and the LFL, TSEs, and LoRA. Table 5 presents the results of an ablation study evaluating the impact of these components across three tasks, by comparing the performance of LightLLM with and without the individual component.

Prompt knowledge and LFL play an important role in enhancing LightLLM’s performance, particularly in unseen scenarios where the system must generalize to new environments not encountered during training. Among the tasks evaluated, localization exhibits the most significant improvement. In the unseen scenario, LightLLM, with these components, achieves a 90th percentile error of 3.46 meters—a substantial reduction from 5.41 meters when they are removed. A similar trend is observed in indoor solar estimation, where prompt knowledge, which includes detailed descriptions of solar cell placement and light conditions, reduces the prediction error. The system achieves an MAPE

of 26.33%, compared to 35.42% without prompt knowledge. Likewise, in the solar forecasting task, CRPS decreases from 2.80 kW to 2.52 kW, further demonstrating the benefits of these enhancements.

TSE are essential for extracting domain-specific features. For instance, in outdoor solar forecasting, the inclusion of a TSE brings a 12% improvement in the CRPS metric, reflecting the encoder’s ability to model temporal dependencies in time-series data. For indoor solar estimation, the TSE contributes to reducing the MAPE from 30.10% to 26.33%.

LoRA ensures efficient fine-tuning by updating task-specific parameters. Across all tasks, removing LoRA leads to consistent performance drops. For example, in indoor solar estimation, MAPE increases from 26.33% to 28.45%, while in indoor localization, the 90th percentile error worsens from 3.46m to 3.79m.

The results in Table 5 clearly demonstrate the unique contributions of each component, which they collectively make LightLLM a versatile and high-performing framework for PLS tasks.

3.4.3 Impact of Knowledge Graph In this section, we evaluate the contribution of the KG mentioned in Section 2.2. In the ablation setting, we remove the KG creation in LightLLM, and instead, using raw data as input. We evaluated the models based on our LSI Localization task.

In our evaluation results, the model with KG achieved significantly lower error, with a median error of 0.98 meters. In contrast, removing the KG led to a 50% decrease in localization accuracy, with the median error increasing to 1.42 meters. The findings validate the design rationale presented in Section 2.2, demonstrating that the KG’s structured data is instrumental in achieving precise localization within the LightLLM framework.

3.4.4 Impact of LLM in LightLLM To assess the importance of the LLM component within LightLLM, we conducted an ablation study by comparing three versions of our model as shown in Figure 12. The purpose of this experiment is to evaluate whether the LLM plays a critical role in the performance

Table 5: Impact of LFL, TSE, and LoRA.

Task	Metric	Configuration			
		Normal	w/o LFL	w/o TSE	w/o LoRA
Outdoor Solar Forecasting	CRPS[kW]↓	2.52	2.80	2.83	2.57
Indoor Localization	90th Percentile Error[m]↓	3.46	5.41	3.73	3.79
Indoor Solar Estimation	MAPE[%]↓	26.33	35.42	30.10	28.45

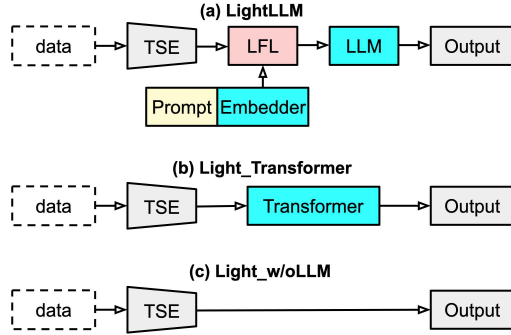


Figure 12: Three ablation versions of LightLLM: LightLLM (full version), Light_Transformer (replacing the LLM component with a single transformer), and Light_w/oLLM (completely removing the LLM, using a simplified model).

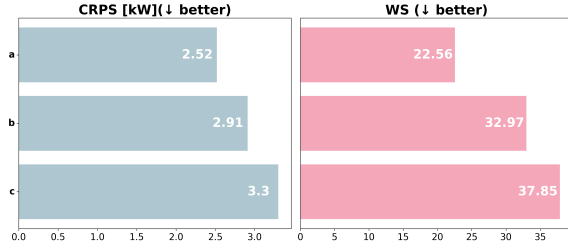


Figure 13: Performance comparison of three ablation versions of LightLLM as shown in Figure 12.

of LightLLM, particularly for tasks like solar energy forecasting. The results in Figure 13 demonstrated that LightLLM (with the LLM component) consistently outperformed the other two versions across all evaluated metrics. This ablation study thus reinforces the importance of incorporating LLMs in LightLLM.

4 FUTURE DIRECTIONS

Exploration of Different LLM Models: We investigated how different pre-trained LLMs impact LightLLM’s performance by comparing LLaMA-7B (7B parameters) with GPT-2 (1.5B parameters) across all tasks. As shown in Figure 14, LLaMA-7B consistently outperformed GPT-2, benefiting from its larger model size and more recent training, which enhance its capacity for reasoning and adaptation to spectral and temporal data.

While our experiments demonstrated the benefits of using larger models, we need to consider the trade-offs between

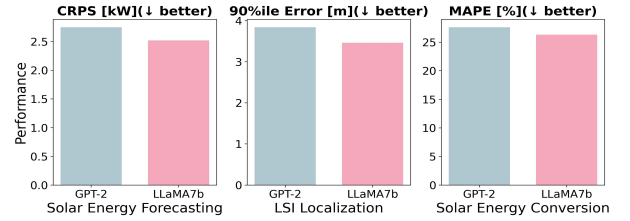


Figure 14: Performance comparison between using different LLM models in LightLLM framework.

model complexity, performance, and resource demands. In certain applications, lighter, open-source models may provide a balance between efficiency and accuracy, making them a feasible option for resource-constrained environments. In contrast, Models with more parameters, like GPT-4, may further boost performance, especially for tasks requiring deeper reasoning. Future research could explore different language model architectures to identify those best suited for various PLS tasks and their impact on adaptability in complex environments.

Task-Specific Tuning Overhead: Although LoRA reduces the need for full model retraining, there is still some overhead involved in task-specific tuning. An important direction for improvement is to enhance its ability to generalize across diverse PLS scenarios. Future research could explore techniques such as multi-task learning or integrate more domain-adaptive pretext tasks, allowing LightLLM to better capture task-specific characteristics, attributes, and structural information.

Knowledge Graph Enhancements: While LightLLM’s KG effectively captures coarse spatial relationships, it does not explicitly model fine-grained physical properties of light, such as reflection, refraction, and scattering. These properties are crucial for high-precision tasks like spectral analysis and fine-grained gesture recognition. A promising direction is to integrate physics-based light propagation models or augment KGs with wavelength-dependent optical properties.

5 RELATED WORK

5.1 PLS Tasks and Algorithms

The field of PLS tasks encompasses applications like indoor localization, solar energy forecasting, and indoor solar energy estimation. These tasks are critical for optimizing energy systems, enhancing smart environments, and improving the efficiency of renewable energy grids.

Indoor localization using light spectral information can significantly improve smart navigation and asset tracking in smart buildings. Light sensor data offers a compelling alternative to traditional localization methods. Device-based VLP systems, such as Pulsar [57], use Angle of Arrival (AoA) sensing to achieve decimeter-level accuracy, but require users to carry dedicated receivers, limiting scalability. Therefore, LightLLM adopts a device-free approach, using spectral sensor data from fixed deployments. Existing device-free models are often environment-specific and rely on fingerprint collection, which demands extensive data gathering and struggles to generalize to new spaces [19, 25, 46].

Solar radiation forecasting helps predict future energy output, improving the reliability and efficiency of energy distribution. Recent advancements in this field have largely focused on deep learning-based methods, such as combining CNN with recurrent neural network (RNN) [17, 36, 42, 58]. There is also a growing trend of replacing RNN with transformers to better capture temporal patterns in the data and improve forecasting accuracy [34].

With the development of efficient solar cells for indoor environments [22, 32, 51], there is a growing need for accurate indoor solar energy estimation. This task focuses on accurately predicting the energy output of photovoltaic systems in indoor conditions to maximize their utility. Current approaches often rely on the solar models that have been presented that use mathematical linear [56] and nonlinear functions [12].

Despite the progress made in solving these tasks, several significant challenges remain. One of the primary issues is the lack of generalization across different environments or unseen data [39]. While data augmentation [16, 59] can help improve performance in the short term, it is computationally expensive and resource-intensive, and it does not fundamentally address the issue of generalization across diverse tasks. Another limitation arises from the fixed nature of these models: a model designed for indoor localization cannot be easily repurposed for solar energy forecasting without re-engineering the entire architecture [49]. This rigidity limits the development of a scalable, multi-task solution capable of handling the diversity of data modalities in PLS tasks.

5.2 Large Language Model and Applications

In recent years, LLMs achieved a great success in the field of natural language processing (NLP). Models like BERT [10], GPT [35], and LLaMA [44], which are built on the Transformer architecture, have demonstrated their ability to perform reasoning, translation and text generation tasks. One of the most significant advancements in LLMs is their ability to learn from massive datasets, capturing latent knowledge that allows them to perform well across diverse tasks.

The flexibility of LLMs has made them increasingly attractive for applications beyond NLP. Techniques like LoRA [18] and prefix-tuning [27] have made it possible to fine-tune LLM for specific downstream tasks with minimal computational overhead. This has led to the exploration of LLM in non-textual domains, including cross-modal learning where text is combined with images, audio, or sensor data.

Recent studies have shown the potential of LLM in multimodal tasks, where they are adapted to handle inputs from various sources beyond text. For example, models like CLIP [40] successfully integrate image-text pairs, allowing LLM to perform visual reasoning tasks. Similarly, PaLM-E [11] has been applied to control robotics tasks, showcasing how LLM can extend their capabilities into non-linguistic domains by integrating structured knowledge from other modalities.

In fields such as healthcare, LLM have been adapted to process clinical text alongside sensor data to provide predictions for medical outcomes. For example, medical models like PaLM-2 [43], MentalLLM [54] have shown success in fine-tuning LLM to accurately interpret clinical notes and predict patient outcomes based on multimodal inputs. Additionally, NetLLM [52] adapts LLM for networking tasks such as view-port prediction and cluster job scheduling by leveraging the feature from networking task data, while TimeLLM [24] specializes in time-series data processing, illustrating the growing versatility of LLM across diverse domains. Similarly, FM-Fi [50] explores LLM applications in human activity recognition (HAR) using RF data. These advancements illustrate the cross-domain adaptability of LLMs, inspiring the development of LightLLM for PLS tasks.

6 CONCLUSION

We presented LightLLM, a unified framework designed to address diverse PLS tasks such as indoor localization, outdoor solar forecasting, and indoor solar estimation. Our extensive evaluations showed that LightLLM outperforms traditional methods and simple prompting approaches, in handling complex, real-world data. Though evaluated on different tasks, LightLLM's design offers a general-purpose solution for other context-rich sensing applications, establishing a foundation for LLM-based frameworks in PLS. Beyond structured predictions, LightLLM also has the potential to be extended for open-ended reasoning tasks, such as generating textual explanations for sensor observations or interpreting spectral and temporal patterns in PLS, which are a key strength of LLMs.

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